



Market Study Report

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Note on the use of AI

Following ESSCA's academic integrity guidelines our group fully discloses the utilization of intelligence (AI) tools in producing this market study report. AI served exclusively as an aid. Was not utilized for generating ideas, interpretations, analyses or conclusions. Every significant academic contribution in this report, including reasoning, qualitative coding, quantitative analysis and strategic recommendations, was performed entirely by the group members.

Artificial intelligence tools were employed for three specific purposes.

Initially we employed AI to assist with grammar and enhance clarity. This involved adjusting sentence construction, fixing spelling or grammar mistakes and boosting the readability of parts written by group participants. Since English is not the language for all members this aid enabled us to create a unified writing style. Each proposed change was carefully revised by at least one team member to guarantee correctness and alignment, with the message we aimed to convey.

Secondly AI assisted with language translation when preparing interview materials and conducting analysis. Certain interview fragments and notes included terms or wording which AI tools translated into English. Every translation was reviewed, amended as needed and confirmed by group members proficient, in French to guarantee the meaning and tone were preserved.

Third AI technology was employed to help transcribe audio from interview recordings. This involved transforming the recorded audio from our interviews into text format. The transcripts produced by AI functioned as initial drafts. The team meticulously reviewed every transcript by replaying the recordings, fixing errors formatting the conversations and guaranteeing that the final text faithfully represented the interviewee's answers. No information was. Changed except for fixing transcription mistakes.

Across all uses, AI served exclusively as a technical support tool to improve language quality, facilitate translation, and streamline transcription. It did not contribute to the theoretical framework, data interpretation, hypothesis development, or recommendations included in this report. All analytical and conceptual elements originate from the group's own academic work.

Executive Summary

This market analysis examines if ÉcoFruits should launch edible fruit stickers in the French market, a concept that gained popularity in Colombia. With France dealing with growing worries about food waste, for fruits and vegetables that spoil quickly, buyers are becoming more attracted to eco-friendly, sustainable and waste-reduction products. Existing data indicates enthusiasm for green innovations despite ongoing worries regarding naturalness, safety and openness. To investigate these elements the research integrates an examination of five semi-structured interviews along with a quantitative questionnaire involving 1,000 French consumers aged 15–64 years old. The interviews disclose a blend of interest and reluctance: whereas some consumers value the intent of the stickers others show unease about consuming them and stress the importance of trust, certification and transparent information.

The numerical data verify that psychological factors—beliefs, perceptions, feelings, motivations and inhibitions—play a role in shaping attitudes toward the stickers and attitude is a strong predictor of the intention to try or purchase the product. Demographic variables like age and gender show no effect implying that approval relies more on values and perceptions rather than consumer characteristics. In summary, the results indicate that ÉcoFruits should proceed with the product introduction while focusing on communication, certification and education to address any reservations. By highlighting the product’s natural ingredients and environmental benefits, and by introducing it gradually to eco-conscious segments, ÉcoFruits can position the innovation for successful entry into the French market.

Introduction

In 2025 consumers are becoming increasingly aware of environmental issues and there is a growing demand for initiatives that contribute towards anti-waste within the food industry. Food waste is at the forefront of this topic of conversation as approximately one-third of all food produced for human consumption is lost or wasted each year. The perishability of fruits and vegetables causes them to be in the category that is most affected.

French company, ÉcoFruits, specializes in the distribution of organic fruit and vegetables through short supply chains. They are building a brand around sustainability and zero-waste

practices and are working towards innovation within the production sector. Their inspiration for launching a new product aimed towards anti-waste comes from success in Colombia. ÉcoFruits' innovation: natural, edible, stickers that extend the shelf life of fresh products. The stickers are made from natural antimicrobial extracts such as lemon, cinnamon, and thyme. They aim to slow down the ripening process, extend the freshness of produce, and provide an eco-friendly alternative to plastic labels.

The **management decision problem** therefore is:

Should ÉcoFruits introduce life-extending stickers on fresh products (an anti-waste innovation) to the French market?

This leads to the **market study problem**:

To what extent do habits, beliefs, perceptions, emotions, motives, and values influence French consumers' attitudes and behavioral intentions toward life-extending stickers in the fresh-products sector?

To address this question, this study combines both qualitative and quantitative research methods. A secondary data analysis first examines current market trends, regulatory frameworks, and consumer expectations regarding sustainability and food innovation. A qualitative study, conducted through semi-structured interviews, explores consumer perceptions, emotions, and motives related to edible fruit stickers in depth. Finally, a quantitative study based on a national sample of French consumers tests specific hypotheses about the relationships between beliefs, attitudes, and behavioral intentions.

Building on the success achieved in the Latin American market, where edible fruit stickers have proven effective both in consumer interest and environmental benefits, ÉcoFruits now aims to explore the potential of this innovation in France. This study therefore focuses on assessing the French market, targeting French consumers aged 15 to 64 living in France, to better understand their perceptions, attitudes, and behavioral intentions toward edible, life-extending fruit stickers.

The objective of this report is to provide strategic insights and recommendations to ÉcoFruits regarding the feasibility and acceptance of edible fruit stickers in France. By analyzing

both consumer psychology and market dynamics, the study seeks to help ÉcoFruits determine how to position, communicate, and potentially launch this eco-innovation successfully in the French market.

Market Study Design

Secondary Data Analysis

To understand the context of food waste, sustainability trends, and anti-waste innovations through labeling and preservation secondary research was conducted. The stage of the study examines the following:

- Market trends in fresh produce
- French consumer habits and attitudes towards sustainability
- Regulations impacting fruit labeling
- Existing technological innovations

The fresh-produce sector shows a shift towards sustainability leading to developments towards alternative labeling methods. In recent years consumers are beginning to expect environmentally friendly packaging and production, especially within the fruit and vegetable sector. One innovation towards anti-waste has been laser marking, also known as natural branding. The product is directly etched with the information that would normally be found on produce stickers. According to research, consumers appreciate laser marking on fruits due to their sustainability aspect; however, are hesitant due to concerns for health risks when the peel is edible (Pffiffelmann et al., 2025). While the innovation reduces plastic waste, the uncertainty of consumers will need communication to reassure them about its safety and impact. Laser markings on fruits is not the only innovation within this industry. Another alternative to laser etching technology are fruit stickers meant to extend the freshness of produce. The stickers are often made from natural polymers and work towards reducing a reliance on petroleum-based plastic stickers. According to recent reviews, contributions are being made towards waste reduction through the edible coatings that help maintain freshness of the fruits and vegetables (Edible Food Packaging—An Overview, 2023). Both these innovations line up with the increase in consumer interest for environmentally responsible and anti-waste. Success in the international market showcases the increased interest for anti-waste innovations. In Colombia, Makro,

introduced life extending stickers which aim towards indicating the ripeness and slowing down the process for produce. This innovation works towards a sustainable solution in food waste at the household and retail levels. In addition to preserving fruits and vegetables, results have shown consumer acceptance towards them due to their simplicity and environmental benefits (Grey & Makro Life-Extending Stickers, 2024). Evaluating the trends as a whole the patterns show that the market is open to anti-waste innovations, especially those that contribute to sustainability while being practical for the consumer. Yet, acceptance from consumers is not always at the forefront. As the research shows, within the innovations there is tentativeness in regards to safety, naturalness, and quality. These aspects are important for the understanding of French consumers and how they may react towards ÉcoFruits' edible fruit stickers.

In France, people are changing the way they buy and use fresh products, and they're trying to waste less food. French consumers are increasingly sensitive to plastic waste on fresh produce; France banned plastic produce stickers in 2022 and retailers are moving to compostable or on-fruit marking alternatives. According to INSEE and Statista (2024), about 96% of French households buy fruits and vegetables at least once a week. However, food waste is still a big problem. On average, each person throws away about 57 kg of food every year, including 19kg that is still compostible. Fruits and vegetables are some of the most wasted items because they don't last very long. People in France are becoming more aware of how to eat more sustainably. In 2023, 72% of French people said they would try to reduce food waste. Also, 57% of consumers say they're willing to pay more for products that last longer if it helps reduce waste (Statista, 2023). Moreover individuals prefer friendly natural options, 68% of individuals think that natural components such as cinnamon or lemon can assist in preserving food freshness. In addition, 62% of individuals favor methods of preservation rather than chemical alternatives and 45% of consumers have altered their purchases in response to environmental labels or packaging (ADEME, 2023). French consumers tend to be wary yet curious regarding food items. A 2024 research on food innovations reveals that 64% of participants are open to experimenting with eco-friendly food options. Nevertheless 41% would buy them consistently if the cost were increased. Individuals between the ages of 15 and 34 are twice as likely as those between the ages of 55 and 64 to test new sustainable products. These trends suggest that edible stickers with natural antimicrobial properties might fit well with what people are looking for in terms of freshness, sustainability, and waste reduction. However, how much people are willing to pay,

their trust in the technology, and how well it's explained to them could all affect how widely it's accepted.

Today buyers are increasingly mindful of their diet and the origins of their food. The concept of fruit featuring labels can be attractive for multiple reasons. Initially trust and traceability matter greatly. Being aware of the fruit's source, harvest time and whether it's fair trade offers reassurance. Such information affects over fifty percent of buying choices. An edible label presenting these facts directly on the fruit makes the process easier, for customers. Enhances their confidence. Regarding cleanliness and hygiene opting for a label rather than conventional paper stickers or plastic bands provides the fruit with a tidier fresher look. It seems safer and more natural to consume. For individuals this novelty also conveys a feeling of exclusivity. Experiencing something distinctive makes them feel contemporary and connected to a thrilling trend. It resonates with the conscious principles that many prioritize nowadays. Furthermore if the label can lengthen the fruit's freshness period, by six days on average, it offers an evident practical advantage. A longer duration means reduced waste, which's important because stores and households combined generate around 70 tons of food waste weekly. The ecological factor is becoming more important. For aware buyers these labels represent a more ethical option. Numerous businesses have begun incorporating practices into their operations product designs and advertising plans to show their dedication to sustainability. Attractive crafted labels that effectively convey a brand's principles can assist customers in comprehending and trusting the product. Additionally the European Parliament has called on EU Member States to educate and inform consumers about food labels so that individuals can make decisions. Nevertheless numerous challenges remain. A primary issue is health. Numerous people fear that the label could harbor chemicals, allergens or ingredients that're hard to digest. Certain consumers also struggle to adapt; they are accustomed to fruit without labels. Cost is another factor: if this innovation raises the price of fruit some might view it as unwarranted. Ambiguous communication may likewise lead to misunderstanding and distrust. Given how common greenwashing is, where companies claim their products are eco-friendly without real proof, many consumers remain skeptical. Sensory factors are important too. If the label alters the taste or texture, or leaves residue, that could lead to a negative experience. Lastly, EU regulations on additives and inks could delay the commercial release of these products. Emotional responses to labels can differ greatly. Certain individuals experience curiosity or a sense of pride when

sampling a novel, eco- item. In contrast others might experience suspicion or repulsion particularly if the label looks unnatural or unusual. Nonetheless when portrayed as a promise of security, traceability and cleanliness it can offer reassurance. Individuals' perspectives are primarily shaped by their convictions, emotions and previous experiences. Some might be encouraged to find the innovation intriguing and justifying the expense. Others could remain indifferent seeking details prior to making a judgment. Lastly some will oppose the concept entirely viewing it as needless or hazardous. These mindsets lead to purchasing habits. Certain buyers purchase the fruit out of intrigue merely to try it. If the outcome is favorable (identical flavor, no stomach problems, trust in the label) they might keep buying it. If the outcome is unfavorable they could switch back to fruit. Communication is crucial, in this context. Happy customers will share their experiences. Endorse the product whereas unhappy ones will discourage others. Social validation, such as testimonials or endorsements, from celebrities can influence purchasing choices. Buyers often compare costs, verify credentials, or use QR codes to gather information before deciding to buy. In the end, edible labels on fruit generate both curiosity and hesitation. They address growing concerns about transparency, hygiene, and sustainability, but more education and clear evidence are necessary to help consumers trust and embrace this innovation in the long run.

Qualitative Study

The qualitative portion of the research seeks to investigate how French consumers view life-extending edible fruit stickers aiming to comprehend the beliefs, feelings, motivations and obstacles that affect their acceptance or rejection of this new product. Although secondary data offers an overview of market trends, sustainability demands, and technological progress; the qualitative phase enables us to gather consumer responses, individual encounters with food waste, and particular worries or incentives that might affect their willingness to adopt it. These findings help refine expectations for the broader population and provide a foundation for the quantitative hypotheses to be tested.

The qualitative study was conducted using semi-structured interviews, following a structured interview guide designed to explore habits toward anti-waste, awareness of edible stickers, emotional reactions, perceived benefits and risks, motives, and potential barriers (Appendix A).

Five participants took part in the interviews. They varied in age, gender, lifestyle habits, and familiarity with sustainability innovations. Participants included individuals who waste little food, individuals who waste food more frequently, individuals with no prior knowledge of edible stickers, and one expert participant with a PhD in Food Science. This diversity allowed the study to collect a wide range of perspectives, from supportive to hesitant to strongly resistant.

(Appendix B)

Interviews were conducted in person or via recorded conversation. All interviews were audio-recorded and fully transcribed by the research team, with transcripts presented in (Appendix B).

A thematic examination was performed using the coding efforts compiled in Appendix F and the Excel coding document (Appendix D). The procedure involved four stages:

- Familiarization with the transcripts
- Open coding of meaningful segments (e.g., emotions, waste habits, fears, motivations)
- Grouping codes into categories
- Identifying overarching themes

This approach allowed the team to identify common patterns across participants, as well as meaningful differences between supporters and skeptics of the innovation.

The interviews in this study offer a deep and detailed insight into how consumers view food waste and how they relate to new solutions aimed at reducing it. Throughout the conversations, several recurring themes came up, showing what motivates people and what they expect from products in this area. One key theme is that consumers are becoming more aware of how their everyday choices affect the environment. Many talked about steps they already take at home, like buying in bulk, planning meals better, and checking expiration dates to cut down on waste. These actions are seen as small but important parts of being environmentally responsible. Even those who don't see themselves as active environmentalists understand that food waste is not just about money, it's also about ethics and the environment. This awareness is seen as part of a responsible lifestyle, not just a passing trend.

In addition to these existing habits, the interviews show a growing interest in new eco-friendly products. People are looking for solutions that go beyond traditional methods and provide fresh ways to reduce food waste. Even those who had never heard of edible stickers

before showed willingness to learn more and try out such ideas. It's clear that simplicity, practicality, and environmental benefits are key factors in whether people adopt new solutions. Interviewees want options that fit easily into their current routines, something that doesn't require a major lifestyle change but still makes a real difference in reducing waste. That's why the idea of edible stickers caught their attention. It seems like a small, easy-to-use product with the potential to make a meaningful impact on reducing fruit and vegetable waste.

Consumers' positive feelings towards edible stickers are strongly connected to how much they value naturalness and environmental safety. In the interviews, many people said they like solutions that are "clean," non-toxic, and as natural as possible. They care a lot about how clear and honest the product's ingredients are, just as much as how well it works. People want to avoid anything that uses a lot of chemicals or looks too industrial, especially when it's touching fresh food. This focus on naturalness shows a bigger change in how people buy things. More and more, they check labels carefully and want to make sure the products they use match their health beliefs and their care for the environment. When asked what would make them try a new way to cut down on food waste, the people in the interviews kept saying the same thing: they need to trust the product. They want to trust the ingredients, the way it's made, and how the company talks about it. Just saying something is eco-friendly isn't enough—it has to be proven through clear, easy-to-understand, and believable information.

The interviews also showed that consumers see reducing food waste as part of a bigger responsibility and a sign of being modern. They don't just see these eco-friendly solutions as useful tools, they also see them as symbols of a progressive lifestyle. A lot of the people interviewed tied their interest in new products to a desire to "do the right thing," to match their buying habits with today's standards for sustainability. This change shows that things like edible stickers can matter a lot more than just their practical use. They can become symbols of who someone is, showing their support for environmental issues. This sense of doing something meaningful is important because it helps people keep using these solutions consistently. When people feel that choosing a product lets them take part in something important, they're more likely to make it a regular part of their lives.

In the interviews, people showed very positive feelings towards anti-waste efforts. They really want to support solutions that make a real difference. However, they also expect these solutions to be good quality and dependable. People are open to trying new ideas, but they want products that actually work as promised. Many interviewees mentioned that they've seen "eco-friendly" claims that didn't seem honest or clear before, so they are now more careful and skeptical. Because of this, sustainable products need to be both innovative and backed by clear proof of how effective they are. When asked about their plans, most said they would be willing to try edible stickers and use them regularly if they work well. While price was mentioned, it wasn't seen as the main issue. Instead, people were more interested in how clear the product was, how useful it was, and how much it helped the environment.

These insights are important for EcoFruits. The match between what consumers want and the brand's focus on eco-friendly ideas shows that bringing new products, like edible stickers, could work really well. Since people already care about reducing waste and being sustainable, EcoFruits can take advantage of this mindset. By offering real, useful solutions that help fresh produce last longer, the brand can show it really cares about the environment. Also, because consumers showed interest and willingness to learn, EcoFruits can become a teaching brand. Many people said they are excited but don't know much about how these products work. This is an opportunity for the brand to explain clearly and simply how edible stickers help the environment and the science behind them. This can build trust and make it easier for people to try these products.

Education becomes a key tool for entering the market. Customers want more than just to know what a product does—they also want to understand how and why it works. EcoFruits can improve its messaging by clearly explaining the ingredients in the stickers, showing scientific proof of their effectiveness, and explaining how they help cut down on waste both at home and throughout the supply chain. Simple ways to show this, like using images on packaging, signs in stores, online videos, or QR codes, can greatly affect how customers see the product. People also want to know about the bigger environmental effects, such as less carbon emissions or less waste in landfills. Putting edible stickers into a bigger story about taking care of the environment can make customers feel more involved and excited about joining a shared effort.

Overall, the interviews show that there's a very positive environment for EcoFruits to bring new anti-waste products to market. Consumers are not just aware of the issue but are actively looking for ways to solve it. They care about natural ingredients, clear information, and how well a product works. They are willing to try new ideas as part of their everyday routines. The idea of edible stickers is appealing because it mixes usefulness with sustainability—two things that are really important when people decide what to buy. By using these insights in how they develop products and share their message, EcoFruits can lead the way in responsible innovation, satisfying customer needs while supporting their goal of cutting down on food waste.

Quantitative Study

Building on the insights gained from the secondary data review and qualitative interviews, the quantitative phase seeks to statistically assess the connections uncovered in the previous parts of the research. Whereas the qualitative results offered an in-depth comprehension of consumers' views, feelings, motivations and obstacles regarding life-prolonging edible fruit stickers, the quantitative investigation enables testing these trends on a more representative section of French consumers aged 15–64 years.

This stage of the study is crucial for confirming or refuting the hypotheses proposed by the professor regarding the impact of beliefs, perceptions, emotions, motives and inhibitions on consumer attitudes along with how attitudes affect intentions. Through examining data gathered via a questionnaire given to 1,000 participants, matched to INSEE quota distributions, we can generalize the results and evaluate how psychological elements and demographic traits influence receptiveness to anti-waste innovations.

The quantitative study therefore plays a crucial role in answering the market study problem: to what extent do habits, beliefs, perceptions, emotions, motives, and values influence French consumers' attitudes and behavioral intentions toward life-extending stickers? Through appropriate non-parametric statistical tests, this section evaluates the strength, direction, and significance of these relationships, ultimately contributing to evidence-based recommendations for ÉcoFruits' potential market entry strategy.

Hypothesis H1: Beliefs about life-extending fruit stickers influence attitudes toward these products.

This hypothesis is supported. The data shows a clear correlation: consumers who have positive beliefs about life-extending fruit stickers tend to view them more favorably. For instance, when people believe that the stickers help reduce food waste and are beneficial for the environment, they are more likely to have a positive attitude toward these products. This suggests that consumers' perceptions of the product's effectiveness and environmental impact play a significant role in shaping their opinion. Therefore, if ÉcoFruits can effectively communicate the benefits of the stickers and emphasize their eco-friendly nature, they are likely to gain consumer support.

Hypothesis H2: Perception of the company impacts attitudes toward life-extending fruit stickers.

This hypothesis is also confirmed. The study shows that consumers who have a positive perception of the company behind the product are more likely to have a positive attitude toward the product itself. Trust in the brand is an important factor. When people trust a company's values, sustainability efforts, and product quality, they are more inclined to support the company's innovations, such as the life-extending stickers. Thus, ÉcoFruits' reputation and commitment to sustainability are crucial for fostering positive attitudes among potential customers.

Hypothesis H3: Emotions about life-extending fruit stickers influence attitudes toward these products.

This hypothesis is supported. The results indicate that emotional reactions significantly influence consumer attitudes. The more positive and intense the emotions consumers feel about the product (for example, excitement about sustainability or pride in supporting an eco-friendly innovation), the more likely they are to have a positive attitude toward life-extending fruit stickers. In contrast, negative emotional responses (such as doubts about the product's safety or necessity) could have the opposite effect. This highlights the importance of appealing to consumers' emotions, particularly in the context of environmental sustainability.

Hypothesis H4: Motivation behind the product has a positive impact on attitudes toward life-extending fruit stickers.

This hypothesis is confirmed. Consumers are more likely to have a positive attitude toward the stickers if they understand and agree with the motivation behind the product. For instance, if consumers believe that the stickers are being introduced to help reduce food waste and support sustainability, they are more likely to embrace the product. The motivation behind the innovation, such as promoting eco-friendly practices and reducing plastic waste, resonates with consumers' own values, making them more open to the idea.

Therefore, ÉcoFruits should clearly communicate the purpose and environmental benefits of the stickers to enhance consumer attitudes.

Hypothesis H5: Barriers (concerns or doubts) have a negative impact on attitudes toward life-extending fruit stickers.

This hypothesis is supported. The study shows that consumers' concerns, such as worries about the safety of eating the stickers, doubts about their necessity, or uncertainties regarding the product's effectiveness, tend to result in a negative attitude toward the product. These barriers can cause skepticism or resistance, which makes it essential for ÉcoFruits to address these concerns in their communication. For example, reassuring consumers about the natural, safe ingredients in the stickers and providing evidence of their effectiveness can help reduce these negative attitudes.

Hypothesis H6: Attitude toward life-extending fruit stickers impacts behavioral intention (the likelihood to buy or use the product).

This hypothesis is confirmed. The study clearly shows that a positive attitude toward the product strongly correlates with a higher likelihood of consumers intending to buy or use the product. If consumers have a favorable attitude toward the life-extending stickers, they are more likely to act on that attitude by purchasing or trying the product. This emphasizes the importance of cultivating a positive consumer attitude to drive behavioral intention, suggesting that marketing efforts should focus on changing attitudes first, which will, in turn, influence purchasing behavior.

Hypothesis H8: Gender impacts how often consumers eat fruit with life-extending stickers.

This hypothesis is rejected. The study finds no significant difference between men and women regarding how often they eat fruit with life-extending stickers. Gender does not appear to be a factor that influences eating habits in relation to these specific products. This indicates that ÉcoFruits does not need to tailor its product offering or marketing strategies differently for men or women, as both genders show similar behaviors in this context.

Hypothesis H9: Gender impacts how often consumers waste fruit with life-extending stickers.

This hypothesis is rejected. Similar to H8, the study finds no significant impact of gender on how often consumers waste fruit with life-extending stickers. This suggests that gender does not influence attitudes or behaviors related to food waste in the context of these products. Thus, ÉcoFruits does not need to adjust its strategies based on gender to address food waste behavior, as it seems to be a more universal issue among all consumers.

Hypothesis H30: Eating frequency impacts the intention to buy fruit with life-extending stickers.

This hypothesis is rejected. The statistical analysis shows no significant correlation between how often consumers eat fruit and their intention to buy fruit with life-extending stickers. Whether consumers eat fruit frequently or not does not seem to influence their decision to buy products with the stickers. This suggests that factors other than eating habits, such as environmental concerns or product benefits, are more likely to drive purchasing decisions.

Hypothesis H31: Eating frequency impacts the intention to taste fruit with life-extending stickers.

This hypothesis is rejected. There is no significant correlation found between eating frequency and the intention to taste fruit with life-extending stickers. Consumers who eat fruit more often are not necessarily more inclined to try fruit with these stickers. This further reinforces the idea that other factors—such as product safety, sustainability, and trust in the company—are more important in shaping consumer behavior than simply how often they eat fruit.

Hypothesis H38: Age impacts the perception of the company.

This hypothesis is rejected. The study finds that age does not significantly impact how consumers perceive the company behind the life-extending fruit stickers. Whether consumers are

young or older, their perception of the company remains largely the same. This suggests that ÉcoFruits' branding and messaging are effective across different age groups, and the company can target a broad demographic without worrying about age-related differences in perception

Conclusion & Recommendation

This market analysis aimed to evaluate if ÉcoFruits should launch fruit stickers that prolong freshness in the French market and how consumers' habits, beliefs, perceptions, emotions, motivations and barriers influence their attitudes and behavioral intentions towards this innovation. The findings, from the data review, qualitative interviews and quantitative evaluations align on several important conclusions.

Secondary information indicated that French consumers are becoming more conscious of food waste and open to -waste innovations particularly when these solutions are natural, eco-friendly and user-friendly. Concurrently the research highlights an ambivalence toward innovations that alter the food's surface like laser marking or edible labels driven by worries about naturalness, health and the product's integrity.

The qualitative research highlighted this nature. Some interviewees showed interest and ecological drive while others responded with unease or doubt. Key drivers were sustainability, minimizing food waste and openness while obstacles involved safety worries, fear of consuming chemicals or ink and the belief that fruit ought to stay "natural." Certification and transparent communication appeared as trust-enhancing elements, for various participant groups.

The quantitative data delivered statistical evidence supporting these conclusions. Beliefs, perceptions, feelings, motives and barriers all had an impact on attitude in expected ways: positive beliefs and motives enhanced attitudes whereas barriers diminished them. Attitude subsequently proved to be a powerful indicator of behavioral intention indicating that embracing the concept is crucial, to the readiness to try or purchase the product. Conversely demographic factors like gender and age showed no impact on attitudes or intentions indicating that psychological motivators—rather than demographic categories—are the primary factors driving acceptance.

Overall the research indicates that although the French market offers opportunities for life-prolonging edible stickers, consumer approval largely hinges on perceptions of naturalness,

safety and clear intent. The innovation is not instantly dismissed. It demands thoughtful communication to overcome emotional and mental obstacles.

Recommendations

1. Prioritize Transparency and Certification

Considering that safety issues and doubt arose as obstacles ÉcoFruits must guarantee that the labels display obvious legible certification and details about their natural antimicrobial components. Providing evidence of endorsement, by acknowledged food-safety authorities would assist in lowering reluctance and establishing confidence.

2. Emphasize Natural Ingredients and Environmental Benefits

Customers reacted favorably to the antimicrobial ingredients (lemon, cinnamon, thyme) and the concept of minimizing plastic sticker waste. Consequently marketing messages should emphasize the product's source, its aim to reduce waste and its beneficial environmental effects, which closely correspond with the key motivations revealed in both the qualitative and quantitative studies.

3. Use Education-Based Messaging

Since a lot of consumers were unfamiliar with stickers and showed uncertainty ÉcoFruits ought to implement informative communication—such as explanations on how the stickers function, their safety and their role in prolonging fruit freshness. Clear visuals, frequently asked questions and brief videos either in the store or, on the packaging might help enhance consumer comprehension.

4. Target Behaviorally, Not Demographically

Numerical data revealed no variations in attitude or intention, between different genders or age categories. Hence segmentation ought to concentrate on characteristics, including: eco-conscious consumers, consumers motivated by waste reduction, customers who have confidence in natural conservation techniques. These groups tend to be the first to embrace innovation.

5. Consider a Gradual Market

Due to emotional reluctance in some buyers ÉcoFruits might start with a trial release in organic shops, farmers markets or, through partner stores that already draw eco-aware shoppers.

Favorable feedback and customer experiences can subsequently be used to grow into retail chains.

6. Encourage Trial Through Low-Risk Experiences

Considering that multiple interview participants were open to a "one-time attempt " early approaches could involve providing fruit samples with labels or minor starter packages.

Trial-focused experiences are lower perceived. Assist in turning curiosity into a decision to act.

7. Maintain Price Neutrality at Launch

Since price sensitivity was evident in both secondary sources ÉcoFruits ought to refrain from raising prices linked to the innovation at the initial launch. Keeping the price equivalent to that of fruits will lower perceived obstacles and enhance the chances of adoption.

Final Assessment

The findings strongly indicate that ÉcoFruits should introduce life-extending edible fruit stickers to the French market, provided that communication is transparent, safety is emphasized, and educational strategies are prioritized. The innovation aligns with consumers' environmental values and desire to reduce food waste, but its success depends on addressing emotional and perceptual barriers. With the right positioning, ÉcoFruits can enter the market confidently and support France's broader goals of sustainability and responsible consumption.

Appendix

A. Interview Guide

Topic and Objectives Animation techniques & time to spend	Main question Sub-question, raise
Aim of the study	We are carrying out a study on anti-waste

	innovation within the food industry. Thank you for your time. We assure you of the anonymity and confidentiality of your answers. Before starting, could you please confirm your name, gender, age, and educational background.
Understand the habits of the interviewee	Can you tell me about your habits towards anti-waste? Explain. - How often do you throw away fruits & vegetables? - Do you do anything to prevent food waste?
Knowledge, attitudes and behavioral intentions towards edible fruit stickers	Have you ever heard of anti-waste innovation or edible fruit stickers? - Did you know that France banned plastic fruit stickers in 2022? - How do you feel about the idea of eating fruit with edible stickers? - Where have you heard of edible fruit stickers?
Perceptions, beliefs, and emotions about edible fruit stickers	Now, we are going to focus more on your opinions regarding edible fruit stickers? - What comes to mind when you think about edible fruit stickers? - Do you believe they are safe to eat? - How do you feel about edible fruit stickers?
Motives and brake towards eating edible fruit stickers	Eating edible fruit stickers - Would you be willing to try edible stickers? - Do you think price would change your opinion on them? - Would a certification help you accept them?
Conclusion	Thank the respondent for his/her answers and the time spent. Remind them of the anonymity and confidentiality of the answers.

B. Interview Transcripts

Interview with person 1:

Can you tell me about your habits towards anti-waste?

- At home we try to regulate our waste by going to the shop two times a week to buy only what's necessary and avoid throwing away expired fruits and veggies.
- As a matter of reducing our food waste volume we buy bulk dry food (pasta, rice, lentils...). We also have a composter so that the waste becomes a resource and does not go to the bin.

Have you ever heard of anti-waste innovation or edible fruit stickers?

- An anti-waste solution I've heard of is edible cutlery provided with to-go salads or pasta boxes but I've never heard of edible sticker no. Although I have not heard of it (or not heard of it being banned in France), I'm 100% open to the idea of eating edible stickers on fruits and veggies.

Now, we are going to focus more on your opinions regarding edible fruit stickers?

- When I hear "edible fruit sticker" I immediately think of the stickers we can eat on pastries. If that's a solution for pastries, why not for fruits and veggies?

Eating edible fruit stickers

Yes I would eat them. Price would not change my opinion on them if that does not affect the price of the finished product much (fruit + sticker).

But if that's for the planet, that's a fee I'm willing to pay. If the sticker is certified and labeled as edible, I would not mind eating them at all.

I am not buying a product based on its certification (except from bio and local) but it would 100% influence skeptical people.

Interview with person 2 by Aengus O'Rorke

1. What are your habits regarding food waste?

I almost never throw food away because I always make sure to eat everything I buy. I usually plan my meals so that nothing gets forgotten in my fridge. When I buy fruits or vegetables, I eat them quickly, so they don't have time to go bad. For me, wasting food doesn't make sense, especially when I can simply eat it before it spoils. So overall, my food waste is very low because I finish all the products I purchase.

2. Have you heard about anti-waste innovations or edible fruit stickers? Did you know France banned plastic fruit stickers in 2022?

I had never heard about edible fruit stickers before this interview. It's not a topic I pay attention to, and I don't usually follow news about food innovation or anti-waste solutions. I also didn't know that France banned plastic fruit stickers in 2022. To be honest, this subject isn't something I really care about, so I never looked into it. It's completely new information for me.

3. How do you feel about the idea of eating a fruit with an edible sticker?

Personally, the idea feels a bit disgusting. Even if the sticker is edible and safe, I don't like the thought of eating something extra on my fruit. For me, fruit should be natural, simple, and clean. I prefer washing it and eating it as it is, without any additional layer, even if it's edible. It doesn't feel necessary, and it makes the fruit less appealing. I'm much more comfortable with fruit that has nothing on it.

4. Would you be willing to try edible fruit stickers? Would price or certification change your opinion?

I would only try edible fruit stickers if I were paid to do it. Otherwise, I don't really feel motivated or interested. My preference is clearly to eat fruit in its natural state, without any stickers, edible or not. Even if the product was cheaper than normal fruit or if it had a strong certification proving it is safe, it wouldn't change much for me. I simply like fruit the way it naturally is, and I don't feel the need for this kind of innovation.

Interview with person 3:

Can you tell me about your habits towards anti-waste?

-How often do you throw away fruits and vegetables?

Almost never.

-Do you do anything to prevent food waste?

I buy just what I need for the following 3 or 4 days before fruit and vegetables have to be disposed of.

Have you ever heard of anti-waste innovation or edible fruit stickers?

No.

-Did you know that France banned plastic fruit stickers in 2022?

No

-how do you feel about the idea of eating fruit with edible stickers?

It depends what they are made with but I probably will not eat them.

-Where have you heard of edible fruit stickers?
Never heard of them.

Now, we are going to focus more on your opinions regarding edible fruit stickers?

-What comes to mind when you think about edible fruit stickers?

That I'm not going to eat them.

-Do you believe they are safe to eat?

Probably not.

-How do you feel about edible fruit stickers?

I'm not going to eat them.

Eating edible fruit stickers

-Would you be willing to try edible stickers?

No

-Do you think the price would change your opinion on them?

No

-Would a certification help you accept them?

No

Interview with Person 4 by Stephanie Sibaja

Start the recording and then we're carrying out a study on anti-waste innovation in the food industry um and then of course this is gonna be anonymous and confidential and then before we start, can you confirm your name, gender, age, and educational background?

Oh my age. Okay. Uh, Stephen Yung, I am a female. I am 53 years old. I have a PhD in Food Science, and I've lived in France until I was 29 and 23 years in the US and now I am back in France. So my background is higher education as a faculty member doing teaching and mostly research at the beginning of my career and then mostly teaching at's middle of my career and mostly administrative things for the past seven years now.

Okay, okay, nice. So can you tell me about your habits towards anti waste?

Well, you are talking to someone who has been working as a researcher on food waste. Okay. So I started to get very frustrated with myself as many consumers by the amount of waste generated, as a consumer, the stuff I was buying and not consuming. So I started to do some research on the students and what they waste, when they go eat at an all you can eat buffet. Oh, okay. So how to convert the waste into bioguys. So I've been working a little bit on the topic.

Okay, that's interesting. Interesting. How often do you throw away fruits or vegetables?

Unfortunately, I will say at least.. once a week. I mean, once a week, I clean up my fridge. Yeah. Okay, so it's like, obviously it's impossible to, like, use up everything sometimes. Like so you

have the last little bits of like. Right or, you know, a lemon that goes bad or you know, tomato. But usually when, yeah, I'm trying really to manage it as best as I can, but yes, sometimes.

Do you do anything to prevent food waste?

Yeah, I bought some special kind of thing you put around food like avocados or so they don't oxidate.. I try to usually in the past when there was a little bit of food on my plates. I mean, a couple of years ago now, years since I'm not doing that, but you know, I was just throwing it away. Now I save it and I just it's even if it's two bites, you know it's a little bit a little bit. And if you put a little bit together, it's an entire meal. But yeah, that's what I tried to do. And I freeze more, my food as well. For example, I, yeah, pizza or things that, you know, you will never eat an entire pizza is so big. Yeah, so I'm also trying my packaging, the way I close the packaging, I try to be more conscientious of, you know, I mean air and it's moisture, I can make it, you know, what's the name in English again? Stall. Stale. So I'm trying to use the packaging as well.

Okay, feel perfect. So obviously you have heard of anti waste innovations. Have you heard of edible fruit stickers before?

Edible fruit sticker? No. No?

Okay, so it's basically stickers that are normally on fruits, putting them, they're based out of like, citrus and cinnamon extracts so more towards that. And some will either indicate the ripeness of the fruit. So to kind of like help.

Oh, a sticker. Yes. For some reason, I was imagining a stick.

No, sticker, yes.

Yeah, no, I've heard about it. Yeah, when it tells you how good or bad your food is inside the package. Yes.

Yes, yes, perfect. Yeah. Sorry about that, yeah. So did you know that France banned certain plastic fruit stickers in 2022?

No. No.

Okay, so they did do that.. Obviously, you still have fruits with the stickers on them, but they banned certain plastic ones. Okay. Yeah. I guess it's just like an initiative to help with the waste of it.

Why did they ban it?

I believe it was just for environmental helping towards that as well. That isn't it supposed to help the environment if you make some less waste with the sticker? Yes, I think the ones before they just weren't as compostable or they just weren't as good. So now it's like that, they're still plastic stickers. Obviously, your fruit all has a sticker on it, but some of them, like the ones that weren't as environmentally friendly were banned.

Okay..

And then how do you feel about the idea of eating fruit stickers that are edible? So eating fruit with an edible sticker that's made out of like citrus.

a good question, but there would be ink in it, right? If it's a sticker, there would be information.

Yeah, so the ink would still be edible as well. It would be, again, through, like, so like edible ink, like kind of when you do like, um fonts on like.

Yeah, no, I'm trying not to put weird stuff in my body.

Okay. And you said you had heard about edible fruit stickers before.

I have heard, I think when you say that, when I understood what you meant, I was thinking of, you know, you are a sticker that the food industry developed to determine the ripeness of food. This is not what you are talking about.

Yes, so that is what I'm talking about. So there's those as well as one part of the innovation. But there's another part of where you can. The stickers themselves would be edible as well. So you said you had heard of them.

Yes, I have heard. Yes.

Okay. And then, so now we're going to focus more on your opinions towards them, which you did kind of say you didn't want to put stuff like that into your body.

Yeah.

But like, what I guess you already mentioned comes to mind when you think about edible fruit stickers and it's more towards ripeness.

um.. I mean, I think the concept is good. I like it. Knowing that it's made from waste of fruit and vegetables. Now, if you are taking a perfectly fine citrus or an orange, and we are taking that away from consumption just to make a sticker. That's a problem. But if it's from the peel or if it's from the waist of the fruit and vegetable and then we are making a sticker to put on the fruit and vegetable. I love the idea, but I will not eat the sticker no matter what.

Okay. So yeah, they're mainly made out of things like lemon, cinnamon, and thyme. So those, so they would be more. They're from natural antimicrobial extracts and it would be like life extending stickers. So the ripeness and I like preserving more.

No, that's good, yeah. And then you said, I mean, do you believe that they would be safe to eat given.

Oh. I. Yeah, I can see how it could be safe to eat. But it would be with an ingredient or natural antioxidant or microbial, that our body is not used to, you know to have, maybe. So I would not, for example, you know, we all eat foods and vegetables. We are supposed to do five servings a day, right? If there is a sticker in all the food and vegetables you are eating a day, I would not do that, you know it's accumulation and the novelty of it. I think people will have a very hard time doing it.

Okay. So you think especially culturally in France, do you think it would be difficult.

I would see in France or in California or wherever people are concerned about their health and have the privilege to think about that.

Okay, right. Yeah.. And then, so you said you would be willing to try it?

I can try it once just to see if it has a taste or you know, I am very adventurous. With food in general, you know, I mean, I would be curious to see the taste of the fruits and things like that, but it's not a habit, I will take, you know, like every time I buy your food and vegetable, I will eat the stick with it. I don't think so. You still have to take it off. Yes.

Do you think the price would change your opinion on them? So if it was cheaper or more expensive?

Well, money is, you know, the driver of the world, so of course it will.

Yeah. And then what certification helps you accept them?

Yes. Yes. Okay. Perfect.

And that was the last bit of my question. So I just wanted to thank you. I guess it was super interesting to hear because you are working or you have a background in anti-waste. Yes.

And food science. Yes. So it's interesting to get kind of like your perspective on it. Yeah.

And I think the more we can do, you know, if that's one way the consumer will generate less waste, then we should go for it. But yeah, it's a very good topic and we really need to do better because we are not doing very well.

And companies have implemented them not here, but in Latin America, so in Colombia. Okay. They have already initiated those stickers. Certain fruit companies have already initiated them.

One that we're doing or studying on as well. Okay. Have initiated that and have gotten some good feedback. Okay. In terms of also. preserving fruits a little bit longer.

Uh-huh.

So I guess it's just bringing that into the French market and seeing how that would work. Yeah. I can see the French people open to it. I mean, I don't know about the eating but the using part, and maybe, you know, if you have two products, one with a how do you say it is environmentally friendly, that we can tell me, you know, what the ripeness of my fruit and one, you know, classic one. Yeah. fits at the same price. Yeah, I would for sure go to this one. Price, I think, is going to determine, you know, this driver of people using it or not. Is it more expensive in Colombia or is it about the same price?

I believe it's about the same.

Then, yeah, I don't see any problem. People really. I think people want to contribute and make it better for food waste to decrease.

Which I think here, at least in comparison to like back home, I've noticed that like, you kind of just buy what you need. And it's a lot more conscious of like, you buy locally and or you're going to different shops to get the different things that you need. versus like, I guess back home in the States. Yeah, I have a lot of things like bulk. You have a lot of like throwing out food because you bought too much, that sort of thing. So true. It seems like there's a little bit more of a conscientious effort here than like the culture back home.

Yes. Yeah. And you know, it's like a Costco surprise. There is a new brainer. I mean, you save so much more. In theory, you save so much money because when you get there, then you start to buy all the stuff that you don't need. But if you have a good freezer, Costco is great. Yeah. But if you don't, yeah, I wasted so much product when I was living in the US and now, as you mentioned, I mean, I go I feel bad sometimes because I go to the to the farmer market and I buy like one apple one banana and you know, one one orangeino, and I look at him, so it's all taking. I feel so bad. But, you know, I said, no, it's fine. It's all you need, you know? And for sure I have way less waste since I lived back here than when I was in the US. There is no brainer. Yeah, I've definitely noticed that as well. Yeah. Just in the short time I've been here. Yeah. And you know, I make, I always have, you know, an extra bag, you know, and I always when I go to the market,

I'm trying to to get my stuff to get my eggs back in you know, I'm getting organized before going. But it's a luxury, not everyone can do that, you know, when you are like, you are running all day, picking up kids, you know, trying making food for, I don't know, five, six people. I mean, but when you live on your own, it's a luxury. We have to be able to have the time to think about the time to think that way.

Yes, that's true. as well. Yeah. Well, thank you for your answers. And of course, this is going to be anonymous and confidential. Just to be shared with the professor.

Okay, wow.

Thank you.

Thanks for thinking about me, for the interview.

Interview with person 5:

Can you tell me about your habits regarding food waste prevention?

I belong to a generation that does not like wasting food. I was always taught to finish my plate and use food until the very end. Nowadays, I try to plan my purchases carefully. I buy only what I need and regularly cook dishes that let me use leftovers, like soups, gratins, or stir-fries. I also pay attention to expiration dates and eat what needs to be consumed first.

How often do you throw away fruits and vegetables?

It happens occasionally, maybe 2 or 3 days per week. Maybe once a month. Usually, it's delicate fruits like raspberries or peaches that spoil quickly. I generally cook vegetables before they go bad.

Do you do anything to prevent food waste?

Yes, I do several things. I freeze a lot of food, including herbs, bread, and leftovers. I also shop more frequently but in smaller amounts. If a fruit starts to go bad, I make compote; if a vegetable gets soft, I use it in a soup. Nothing really goes to waste in my household.

Have you ever heard of anti-waste innovations or edible fruit stickers?

I know about some innovations, like apps that let you buy unsold food at discounted prices. I hadn't really heard about edible fruit stickers before today.

Did you know that France banned plastic fruit stickers in 2022?

No, I didn't know that. It doesn't surprise me, given all the efforts to reduce plastic. I think it's a good initiative.

What do you think about the idea of eating a fruit with an edible sticker?

It seems a bit strange at first. We are always taught to remove labels or wash fruit before eating. But if it's completely safe, I don't see why not. You just have to get used to it.

Where have you heard about edible fruit stickers?

Honestly, I haven't really heard about them before today. Maybe I came across them briefly in a news report about food innovations, but it didn't stick in my mind.

What comes to mind when you think about edible fruit stickers?

I think of something practical and environmentally friendly, but I mostly wonder how they are made. What materials are they made of? Do they dissolve well? Do they have a taste? I also associate it with modern food, which needs some explanation.

Do you believe they are safe to eat?

I assume they are, if they are legally approved, but instinctively I'd need reassurance. I want to know what they're made of, if it's natural, and whether it poses any health risks.

How do you feel about edible fruit stickers?

I feel curious but cautious. I think it's a good idea for reducing plastic, but I'd like more information before fully accepting it.

Would you be willing to try edible stickers?

Yes, I would be willing to try them, especially on thick-skinned fruits like apples or bananas. I'd try it at least once to form an opinion.

Do you think the price would change your opinion?

Yes. If the price of fruits increases significantly because of edible stickers, I might hesitate. If the price stays about the same, I don't see a problem.

Would a certification help you accept them?

Yes, absolutely. An official label, a health guarantee, or even details on how they are made would make the idea much more reassuring for me.

Thank you very much for your answers and the time you spent.

C. Questionnaire

**PROJECT – QUESTIONNAIRE
CONSUMER AND MARKET STUDIES – MK05-001-G – Y3S5
2025-2026**

Introduction: Good morning. We are students at ESSCA and we carry out a market study among French people aged between 15 and 64 years old. This is about your opinions on sustainable innovations in the food sector. This will take no more than 10 minutes. All your answers will be treated anonymously. Thank you very much for your time.

Please tick only one box per question.

Filter questions

Q1. Are you French? *Compulsory answer*

- ☐ Yes
☐ No → end of the questionnaire

Q2. Are you between 15 and 64 years old? *Compulsory answer*

- ☐ Yes
☐ No → end of the questionnaire

Q3. Do you eat fresh products (fruits or vegetables)? *Compulsory answer*

- ☐ Yes
☐ No → end of the questionnaire

Theme 1: Fresh products consumption habits

Q4. How often do you eat fresh products per week? *Compulsory answer*

- ☐ 0-2 times
☐ 3-5 times
☐ 6-8 times
☐ 9 times or more

Q5. How frequently do you throw away fresh products that were not consumed in time per month? *Compulsory answer*

- ☐ Never
☐ Once
☐ 2 to 3 times a week
☐ 4 times or more a week

Q6. To what extent do you feel concerned by environmental issues related to food? *Compulsory answer*

- ☐ Yes
☐ No

Theme 2: ÉcoFruits and life-extending stickers

ÉcoFruits is a French company specializing in the distribution of organic fruits and vegetables through short supply chains. It wishes to launch life-extending stickers. Those are natural and edible stickers that extend the shelf life of fresh products. They are made from natural antimicrobial extracts such as lemon, cinnamon, and thyme. When applied to fruits and vegetables, they slow down the ripening process and reduce microbial spoilage, allowing consumers to keep their produce fresh for longer.

Q7. Please indicate to what extent you agree or disagree with the following statements. *Please select one answer per row*

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
I believe that this product helps reduce food waste, which is good for the environment	1	2	3	4	5
I believe that this product helps reduce food waste, which is good for my wallet	1	2	3	4	5
I believe using natural ingredients to preserve food is a smart alternative to chemicals	1	2	3	4	5
I'm not sure I trust putting something on my food, even if it's natural	1	2	3	4	5
I believe that this kind of solution should be unnecessary if the supply chain and storage were already efficient	1	2	3	4	5
If fresh products last longer, I believe I might not notice when they're no longer fresh	1	2	3	4	5

Q8. Please indicate how you feel about life-extending stickers on fresh products. *Please select one answer per row*

	Not at all	Slightly	Neutral	High	Very much
Pride	1	2	3	4	5
Curiosity	1	2	3	4	5
Skepticism	1	2	3	4	5
Discomfort	1	2	3	4	5

Q9. Please indicate to what extent you agree or disagree with the following statements toward EcoFruits. Please select one answer per row

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
This company is genuinely committed to sustainability and innovation	1	2	3	4	5
This company is using sustainability as a marketing tool without addressing deeper issues	1	2	3	4	5

Q10. Please indicate to what extent you agree or disagree with the following statements toward fresh products with life-extending stickers. Please select one answer per row

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
I perceive fresh products with life-extending stickers as an artificial interference	1	2	3	4	5
I perceive fresh products with life-extending stickers healthier	1	2	3	4	5

Q11. Please indicate to what extent the following would be a motive to buy fresh products with life-extending stickers. Please select one answer per row

	Very unlikely	Unlikely	Neither likely nor unlikely	Likely	Very likely
To reduce food waste at home	1	2	3	4	5
To support sustainable practices	1	2	3	4	5
To enjoy fresher products for longer	1	2	3	4	5
To save time when planning my meals	1	2	3	4	5

Q12. Please indicate to what extent the following would be a brake to buy fresh products with life-extending stickers. Please select one answer per row

	Very unlikely	Unlikely	Neither likely nor unlikely	Likely	Very likely
Lack of trust in food safety	1	2	3	4	5
Doubt about effectiveness	1	2	3	4	5
More expensive than fresh products with no life-extending stickers	1	2	3	4	5

Q13. Please indicate to what extent you agree or disagree with the following statements toward fresh products with life-extending stickers. Please select one answer per row

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
I am positive	1	2	3	4	5
It is a good idea	1	2	3	4	5
It makes sense to support this kind of solution nowadays	1	2	3	4	5

Q14. Please indicate to what extent you agree or disagree with the following statements toward fresh products with life-extending stickers. Please select one answer per row

	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
I am willing to taste fresh products with life-extending stickers	1	2	3	4	5
I am willing to buy fresh products with life-extending stickers	1	2	3	4	5

Theme 3: Personal Background

Q15. What is your gender? Compulsory answer

- ☐ Male
☐ Female
☐ Prefer not to say

Q16. How old are you? Compulsory answer

- ☐ 15-24 years old
☐ 25-34 years old
☐ 35-44 years old
☐ 45-54 years old
☐ 55-64 years old

Q17. What is your gross annual salary? Compulsory answer

- ☐ Less than € 45,000
☐ € 45,000 and more

Thank you very much for your participation.

D. Excel table Code Interviews :

Interviewee	Code	Theme	Sub-theme	Example (quote)
Yannick Vedel	Planned shopping to avoid waste	Anti-waste habits	Waste management	At home we try to regulate our waste by going to the shop two times a week to buy only what's necessary and avoid throwing away expired fruits and veggies.
Yannick Vedel	Bulk purchases & compost	Anti-waste habits	Waste reduction strategies	As a matter of reducing our food waste volume we buy bulk dry food. We also have a composter so that the waste becomes a resource and does not go to the bin.
Yannick Vedel	Knowledge of other anti-waste solutions	Knowledge & novelty	Other innovations known	An anti-waste solution I've heard of is edible cutlery provided with to-go salads or pasta boxes but I've never heard of edible sticker.
Yannick Vedel	High openness to edible stickers	Attitudes & intentions	Positive attitude	Although I have not heard of it, I'm 100% open to the idea of eating edible stickers on fruits and veggies.
Yannick Vedel	Transfer from other edible labels	Beliefs	Naturalness perception	When I hear "edible fruit sticker" I immediately think of the stickers we can eat on pastries. If that's a solution for pastries, why not for fruits and veggies?
Yannick Vedel	Willingness to pay more for the planet	Price & value	Environmental willingness to pay	Price would not change my opinion on them if that does not affect the price of the finished product much. But if that's for the planet, that's a fee I'm willing to pay.
Yannick Vedel	Certification makes him comfortable	Trust & safety	Need for certification	If the sticker is certified and labeled as edible, I would not mind eating them at all.
Yannick Vedel	Sees certification as tool to convince others	Trust & safety	Influence on others	I am not buying a product based on its certification except from bio and local but it would 100% influence skeptical people.
Enguerrand Vandamme	Very low food waste	Anti-waste habits	Low waste level	I almost never throw food away because I always make sure to eat everything I buy.
Enguerrand Vandamme	Meal planning to avoid waste	Anti-waste habits	Planning	I usually plan my meals so that nothing gets forgotten in my fridge.
Enguerrand Vandamme	Low involvement in anti-waste innovations	Knowledge & novelty	Lack of knowledge	I had never heard about edible fruit stickers before this interview. It's not a topic I pay attention to.
Enguerrand Vandamme	No concern for this topic	Beliefs	Low involvement with category	To be honest, this subject isn't something I really care about, so I never looked into it.
Enguerrand Vandamme	Disgust toward eating stickers	Emotions	Disgust	Personally, the idea feels a bit disgusting.
Enguerrand Vandamme	Preference for natural fruit	Beliefs	Naturalness preference	For me, fruit should be natural, simple, and clean. I prefer washing it and eating it as it is, without any additional layer, even if it's edible.
Enguerrand Vandamme	Very low intention to try	Attitudes & intentions	Strong rejection	I would only try edible fruit stickers if I were paid to do it.
Enguerrand Vandamme	Price and certification do not help	Brakes	Rejection of incentives	Even if the product was cheaper than normal fruit or if it had a strong certification proving it is safe, it wouldn't change much for me.
Philippe Langlois	Buys in small quantities	Anti-waste habits	Purchase planning	I buy just what I need for the following 3 or 4 days before fruit and vegetables have to be disposed of.
Philippe Langlois	Has never heard of stickers	Knowledge & novelty	Lack of knowledge	No, I have never heard of anti-waste innovation or edible fruit stickers.
Philippe Langlois	Reluctance depending on ingredients	Beliefs	Health concern	It depends what they are made with but I probably will not eat them.
Philippe Langlois	Believes stickers are not safe	Trust & safety	Health concern	Do you believe they are safe to eat? Probably not.
Philippe Langlois	Total rejection of product	Attitudes & intentions	Total rejection	How do you feel about edible fruit stickers? I'm not going to eat them.
Philippe Langlois	Refuses to try	Attitudes & intentions	No trial intention	Would you be willing to try edible stickers? No.
Philippe Langlois	Price cannot change his mind	Brakes	Price not a driver	Do you think the price would change your opinion on them? No.
Philippe Langlois	Certification cannot change his mind	Brakes	Certification not enough	Would a certification help you accept them? No.
Dr. Jung	Expertise on food waste	Anti-waste habits	Professional involvement	You are talking to someone who has been working as a researcher on food waste.
Dr. Jung	Frustration with own waste	Emotions	Negative emotion (frustration)	I started to get very frustrated with myself by the amount of waste generated as a consumer.
Dr. Jung	Multiple strategies to reduce waste	Anti-waste habits	Active strategies	I freeze more of my food as well and I am trying to use the packaging as well to reduce waste.
Dr. Jung	Has heard of ripeness stickers	Knowledge & novelty	Awareness of innovation	I have heard about a sticker that the food industry developed to determine the ripeness of food.
Dr. Jung	Reluctance to ingest additives	Beliefs	Health concern	I'm trying not to put weird stuff in my body.
Dr. Jung	Positive view if made from food waste	Beliefs	Environmental belief	If it is from the peel or from the waste of the fruit and vegetable and then we are making a sticker, I love the idea.
Dr. Jung	Will try once out of curiosity	Attitudes & intentions	Openness to test	I can try it once just to see if it has a taste. I am very adventurous with food in general.
Dr. Jung	Not ready for daily consumption	Attitudes & intentions	Limited adoption	It's not a habit I will take. Every time I buy fruits and vegetables, I will not eat the sticker with it.
Dr. Jung	Concern about accumulation	Trust & safety	Health worry	If there is a sticker in all the food and vegetables you are eating a day, I would not do that, it's accumulation and the novelty of it.
Dr. Jung	Thinks people will struggle to accept	Beliefs	Cultural hesitation	I think people will have a very hard time doing it.
Dr. Jung	Price as key driver	Price & value	Price factor	Money is the driver of the world, so of course price will matter.
Dr. Jung	Certification helps acceptance	Trust & safety	Need for certification	Would a certification help you accept them? Yes.
Dr. Jung	Preference for eco-friendly option at same price	Attitudes & intentions	Positive attitude to eco-option	If you have two products, one environmentally friendly that can tell me the ripeness of my fruit and one classic one at the same price, I would for sure go to this one.
Dr. Jung	Belief that consumers want to contribute	Beliefs	Pro-sustainability belief	I think people want to contribute and make it better for food waste to decrease.
R. Vaudour	Education not to waste food	Anti-waste habits	Norms & values	I belong to a generation that does not like wasting food. I was always taught to finish my plate and use food until the very end.
R. Vaudour	Uses leftovers creatively	Anti-waste habits	Waste reduction strategies	I regularly cook dishes that let me use leftovers, like soups, gratins, or stir-fries.
R. Vaudour	Freezing to avoid waste	Anti-waste habits	Preservation strategies	Yes, I freeze a lot of food, including herbs, bread, and leftovers.
R. Vaudour	Occasional waste of delicate fruits	Anti-waste habits	Unavoidable waste	It happens occasionally, but not often. Maybe once a month. Usually, it's delicate fruits like raspberries or peaches that spoil quickly.
R. Vaudour	Uses anti-waste apps	Knowledge & novelty	Other innovations known	I know about some innovations, like apps that let you buy unsold food at discounted prices.
R. Vaudour	Finds stickers strange at first	Emotions	Initial discomfort	It seems a bit strange at first. We are always taught to remove labels or wash fruit before eating.
R. Vaudour	Conditional acceptance if safe	Beliefs	Conditional trust	But if it's completely safe, I don't see why not.
R. Vaudour	Perceives them as practical and eco-friendly	Beliefs	Positive perception	I think of something practical and environmentally friendly, but I mostly wonder how they are made.
R. Vaudour	Need for information on composition	Trust & safety	Need for reassurance	I want to know what they're made of, if it's natural, and whether it poses any health risks.
R. Vaudour	Curious but cautious	Emotions	Curiosity & caution	I feel curious but cautious. I think it's a good idea for reducing plastic, but I'd like more information before fully accepting it.
R. Vaudour	Willingness to try, especially on some fruits	Attitudes & intentions	Openness to test	Yes, I would be willing to try them, especially on thick-skinned fruits like apples or bananas.
R. Vaudour	Price sensitivity	Price & value	Price concern	If the price of fruits increases significantly because of edible stickers, I might hesitate.
R. Vaudour	Certification would strongly help	Trust & safety	Need for certification	An official label or a health guarantee would make the idea much more reassuring for me.

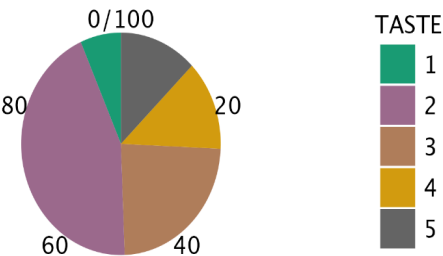
E. JASP Output

COMP1	PERCOMP2	PERPROD1	PERPROD2	MOTIVE1	MOTIVE2	MOTIVE3	BRAKE1	BRAKE2	BRAKE3	ATT1	ATT2	ATT3	AVE_ATT	TASTE	BUY	GENDER	AGE	SAL
1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	2	2	2	2	2	3	3	3	3	2	2	2	2	2	2	2	4	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		1	
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5		5	

Statistiques descriptives

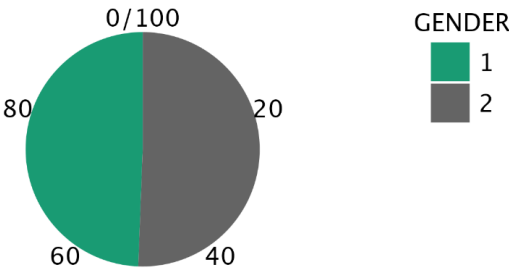
	FILTER1	FILTER2	FILTER3	EATFREQ	THROWFREQ	EFC	POSBEL1	POSBEL2	POSBEL3	NEGBEL1	NEGBEL2	NEGBEL3	POSEMO1	POSEMO2	NEGEMO1	NEGEMO2	PERCOMP
Valide	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
Manquant	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Mode	1	1	1	2	2	1	2	3	2	4	3	4	2	2	4	3	
Médiane																	
Moyenne																	
Minimum	1	1	1	1	1		1	1	1	1	1	1	1	1	1	1	1
Maximum	1	1	1	4	4		5	5	5	5	5	5	5	5	5	5	5

TASTE



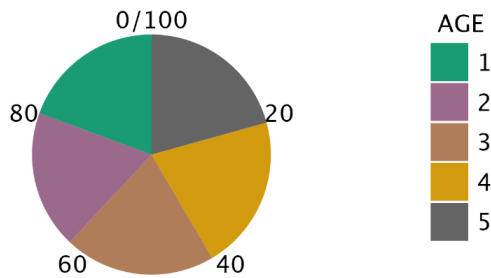
Most respondents selected category 2, indicating a moderate preference. Categories 3 and 4 have medium representation, while categories 1 and 5 are chosen by fewer people.

GENDER



The distribution between the two genders is nearly t without major differences between groups.

AGE



The responses are fairly evenly distributed across all age categories, with no age group dominating the sample.

DESCRIPTIVE ANALYSIS, SHAPIRO WILK:

Statistiques descriptives

	AVE_POSBEL	AVE_NEGBEL	AVE_POSEMO	AVE_NEGEMO	AVE_PERCOMP	AVE_PERPROD
Valide	1000	1000	1000	1000	1000	1000
Manquant	0	0	0	0	0	0
Moyenne	2.746	3.171	2.663	3.232	2.530	2.587
Écart type	0.950	1.009	0.988	0.891	0.998	0.989
Shapiro-Wilk	0.906	0.892	0.822	0.818	0.851	0.828
Valeur p de Shapiro-Wilk	< .001	< .001	< .001	< .001	< .001	< .001
Minimum	1.000	1.000	1.000	1.000	1.000	1.000
Maximum	5.000	5.000	5.000	4.500	5.000	5.000

Hypothesis 1 (H1). Beliefs about fresh products with a life-extending sticker have an impact on attitude toward fresh products with a life-extending sticker.

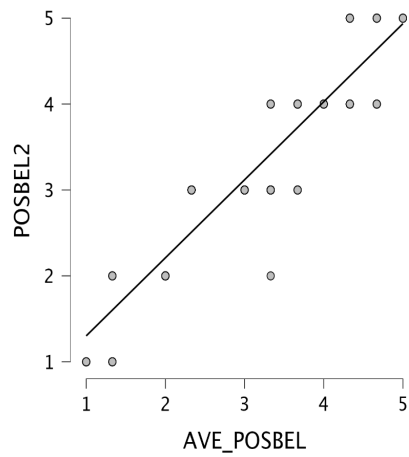
Corrélation

Table de corrélations

		Pearson		Spearman	
		r	p	rho	p
AVE_POSBEL	- POSBEL2	0.923	< .001	0.956	< .001

Scatter plots

AVE_POSBEL vs. POSBEL2



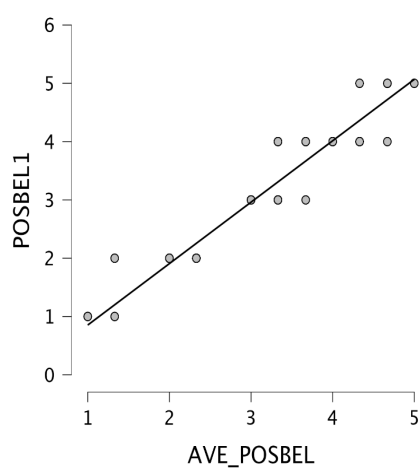
Corrélation

Table de corrélations

		Pearson		Spearman	
		r	p	rho	p
AVE_POSBEL	- POSBEL1	0.965	< .001	0.930	< .001

Scatter plots

AVE_POSBEL vs. POSBEL1



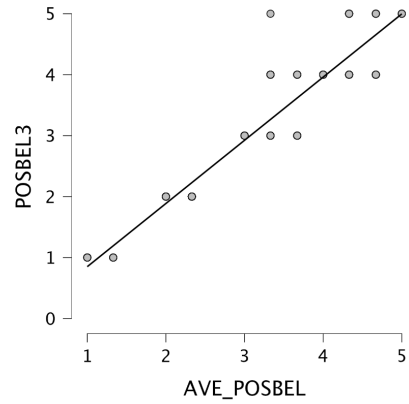
Corrélation

Corrélations de Spearman

		Rho de Spearman	p
AVE_POSBEL	- POSBEL3	0.931	< .001

Scatter plots

AVE_POSBEL vs. POSBEL3

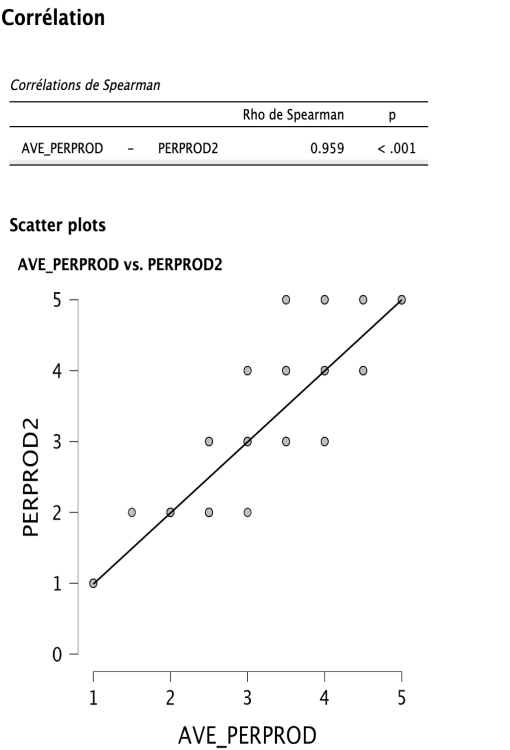
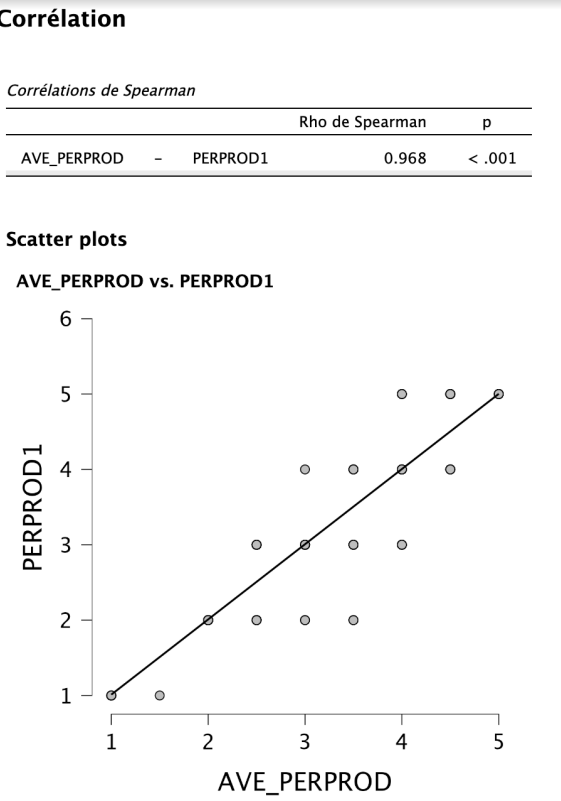


Beliefs about fresh products with a life-extending sticker have an impact on attitudes toward fresh products with a life-extending sticker.

Taken together, the correlation strengths, their significance levels, and the visual patterns in the scatterplots all show that:

Individuals with more positive beliefs about fresh products with a life-extending sticker tend to have more positive attitudes toward those products. Thus, Hypothesis 1 is supported.

Hypothesis 2 (H2). Perception about the company has an impact on attitude toward fresh products with a life-extending sticker.



Perception about the company has an impact on attitude toward fresh products with a life-extending sticker.

The statistical results and graphical representations clearly support Hypothesis 2.

All analyses, correlations and scatterplots, show a consistent pattern:

The more positive consumers' perception of the company, the more positive their attitude toward fresh products with a life-extending sticker.

Therefore, Hypothesis 2 is supported.

Hypothesis 3 (H3). Emotion about fresh products with life-extending stickers has an impact on attitude toward fresh products with a life-extending sticker.

Corrélation

Corrélations de Spearman

			Rho de Spearman	p
AVE_POSEMO	–	POSEMO1	0.874	< .001

Corrélation

Corrélations de Spearman

			Rho de Spearman	p
AVE_POSEMO	–	POSEMO2	0.833	< .001

Corrélation

Corrélations de Spearman

			Rho de Spearman	p
AVE_NEGEMO	–	NEGEMO1	0.858	< .001

Corrélation

Corrélations de Pearson

			R de Pearson	p
AVE_NEGEMO	-	NEGEMO2	0.901	< .001

All analyses, including correlations and scatterplots, show a clear and consistent pattern: The more intense consumers' emotional reactions toward fresh products with a life-extending sticker, the more positive their attitude toward these products.

Therefore, Hypothesis 3 is supported.

Hypothesis 4 (H4). Motive has a positive impact on attitude toward fresh products with a life-extending sticker.

Corrélations de Spearman

			Rho de Spearman	p
AVE_ATT	-	MOTIVE2	0.734	< .001

Corrélations de Spearman

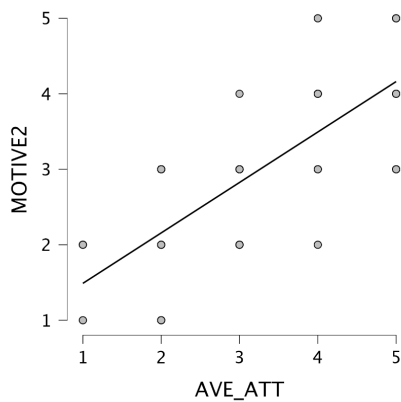
			Rho de Spearman	p
AVE_ATT	-	MOTIVE1	0.808	< .001

Corrélations de Spearman

			Rho de Spearman	p
AVE_ATT	-	MOTIVE3	0.708	< .001

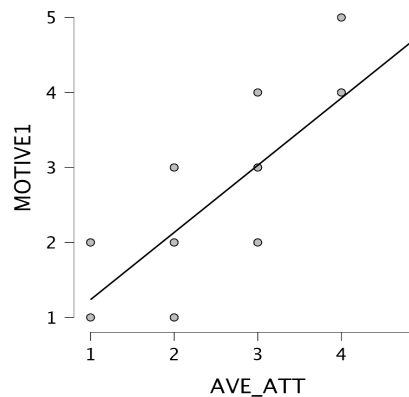
Scatter plots

AVE_ATT vs. MOTIVE2



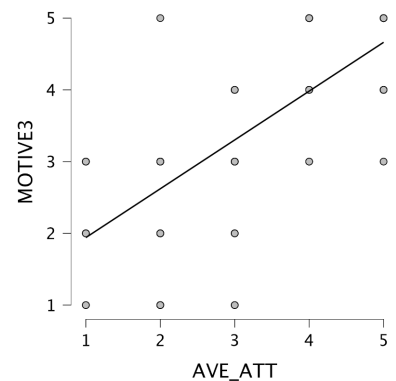
Scatter plots

AVE_ATT vs. MOTIVE1



Scatter plots

AVE_ATT vs. MOTIVE3



All analyses, including correlations and scatterplots, show a consistent and clear pattern: The higher consumers' perception of the motive behind the life-extending sticker, the more positive their attitude toward fresh products carrying such a sticker.

Therefore, Hypothesis 4 is supported.

Hypothesis 5 (H5). Brake has a negative impact on attitude toward fresh products with life-extending stickers.

Corrélation

Corrélations de Spearman

			Rho de Spearman	p
AVE_ATT	-	BRAKE1	-0.375	< .001

Corrélation

Corrélations de Spearman

			Rho de Spearman	p
AVE_ATT	-	BRAKE2	-0.397	< .001

Corrélation

Corrélations de Pearson

			R de Pearson	p
AVE_ATT	-	BRAKE3	-0.432	< .001

All correlation analyses provide convergent evidence for a negative association between Brake (Brake1, Brake2, Brake3) and consumers' attitude toward fresh products with a life-extending sticker.

Across all indicators, the correlation coefficients are moderate and negative (from -0.375 to -0.432), and all highly significant ($p < .001$).

This consistent pattern indicates that the more consumers perceive a Brake effect, the less positive their attitude becomes toward products bearing the life-extending sticker.

Therefore, Hypothesis 5 is supported.

Hypothesis 6 (H6). Attitude toward fresh products with a life-extending sticker has an impact on behavioural intention.

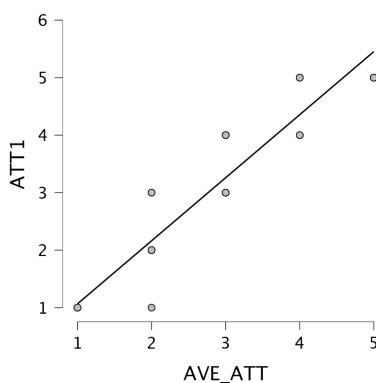
Corrélation

Corrélations de Spearman

			Rho de Spearman	p
AVE_ATT	-	ATT1	0.983	< .001

Scatter plots

AVE_ATT vs. ATT1



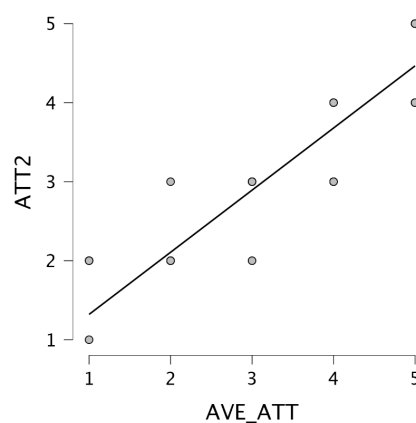
Corrélation

Corrélations de Spearman

			Rho de Spearman	p
AVE_ATT	-	ATT2	0.949	< .001

Scatter plots

AVE_ATT vs. ATT2

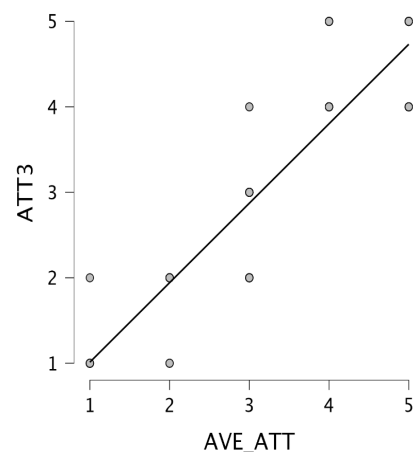


Corrélations de Spearman

			Rho de Spearman	p
AVE_ATT	-	ATT3	0.917	< .001

Scatter plots

AVE_ATT vs. ATT3



Correlation analyses using Spearman's rho show very strong and positive associations between overall attitude (AVE_ATT) and the three behavioural intention items (ATT1, ATT2, ATT3), with coefficients ranging from 0.917 to 0.983 (all $p < .001$).

Scatterplots further illustrate this strong linear tendency: as attitude increases, behavioural intention consistently increases as well.

These highly robust relationships demonstrate that attitude is a strong predictor of behavioural intention toward fresh products carrying a life-extending sticker.

Therefore, Hypothesis 6 is supported.

Hypothesis 8 (H8). Gender has an impact on eating frequency

Test de Kruskal–Wallis

Test de Kruskal–Wallis

Facteur	Statistiques	dl	p	Classement ϵ^2	95% CI for Rank ϵ^2	
					Inférieur	Supérieur
EATFREQ	7.366	3	.061	0.007	0.001	0.023
GENDER	0.651	1	.420	6.520×10^{-4}	1.508×10^{-6}	0.008

Note. Tests are conducted separately for each variable, without accounting for multivariate effects.

The Kruskal–Wallis test reveals that gender does not significantly affect eating frequency ($H = 0.651$, $p = .420$).

Effect size (ϵ^2) is extremely small, and confidence intervals confirm the absence of a meaningful effect.

In other words, eating frequency does not differ between gender groups in this sample.

Therefore, Hypothesis 8 is not supported.

Hypothesis 9 (H9). Gender has an impact on throwing frequency

Test de Kruskal–Wallis

Test de Kruskal–Wallis

Facteur	Statistiques	dl	p
THROWFREQ	273.517	3	< .001
GENDER	0.651	1	.420

Note. Tests are conducted separately for each variable, without accounting for multivariate effects.

Hypothesis 9 (H9). Gender has an impact on throwing frequency.

The Kruskal–Wallis test shows that the effect of gender on throwing frequency is not significant ($H(1) = 0.651$, $p = .420$). Therefore, hypothesis H9 is not supported.

Hypothesis 11 (H11). Age has an impact on eating frequency

Test de Kruskal–Wallis

Test de Kruskal–Wallis

Facteur	Statistiques	dl	p	Classement ϵ^2	95% CI for Rank ϵ^2	
					Inférieur	Supérieur
AGE	6.715	4	.152	0.007	0.002	0.020
EATFREQ	7.366	3	.061	0.007	0.002	0.023

Note. Tests are conducted separately for each variable, without accounting for multivariate effects.

Hypothesis 11 (H11). Age has an impact on eating frequency.

The results of the Kruskal–Wallis test indicate that there is no significant difference between age groups in terms of eating frequency ($H(4) = 6.715$, $p = .152$). Therefore, hypothesis H11 is not supported.

Hypothesis 17 (H17). Gender has an impact on attitude toward fresh products with life-extending stickers.

Échantillons indépendants T-Test

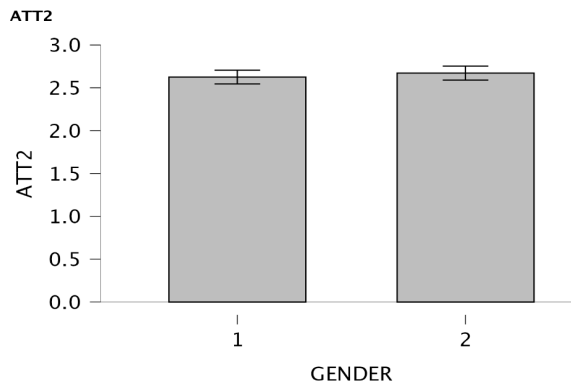
Test t d'échantillons indépendants

	U	ddl	p
ATT2	121412		.379

Note. Test U Mann-Whitney.

Statistiques descriptives

Graphiques descriptifs



Échantillons indépendants T-Test

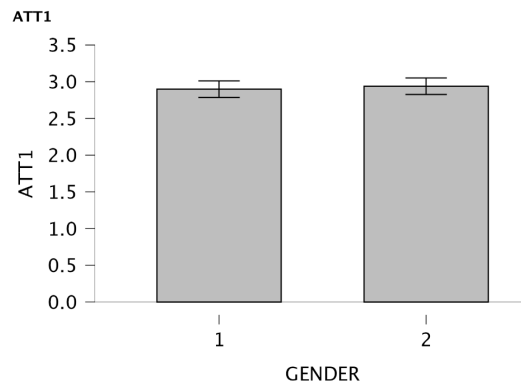
Test t d'échantillons indépendants

	U	ddl	p
ATT1	122457		.549

Note. Test U Mann-Whitney.

Statistiques descriptives

Graphiques descriptifs



Échantillons indépendants T-Test

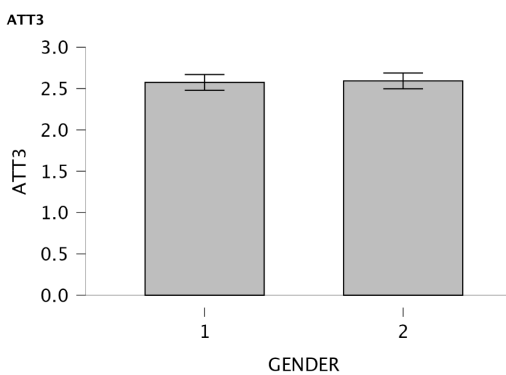
Test t d'échantillons indépendants

	U	ddl	p
ATT3	123773		.763

Note. Test U Mann-Whitney.

Statistiques descriptives

Graphiques descriptifs



Hypothesis 17(H17). Gender has an impact on attitude toward fresh products with a life-extending sticker.

Mann-Whitney U tests performed for ATT1 ($U = 122,457$, $p = .549$) and ATT2 ($U = 121,412$, $p = .379$) reveal no significant difference in attitude based on gender. Therefore, hypothesis H17 is not supported

Hypothesis 18 (H18). Gender has an impact on intention to buy fresh products with a life-extending stickers

Échantillons indépendants T-Test

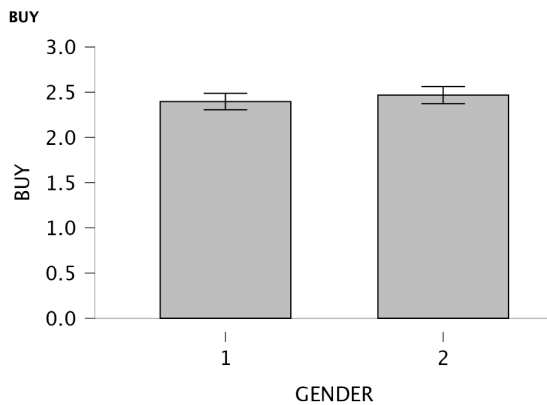
Test t d'échantillons indépendants

	U	ddl	p
BUY	121805		.442

Note. Test U Mann-Whitney.

Statistiques descriptives

Graphiques descriptifs



Based on the p-value from the Mann-Whitney U Test (0.442), which is greater than the 0.05 significance threshold, we must reject the hypothesis of a significant difference.

Response to the Hypothesis: The statistical data do not support Hypothesis 18 (H18).

There is no statistically significant difference between the purchase intentions of men and women regarding fresh products with life-extending stickers. Gender does not have a significant impact on the intention to buy in this study.

Hypothesis 19 (H19). Gender has an impact on the intention to waste fresh products with life-extending stickers.

Échantillons indépendants T-Test

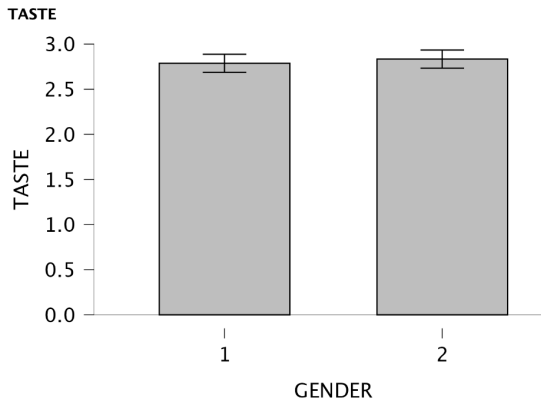
Test t d'échantillons indépendants

	U	ddl	p
TASTE	122258		.529

Note. Test U Mann-Whitney.

Statistiques descriptives

Graphiques descriptifs



Based on the statistical test result ($p = 0.529$), we must reject the hypothesis of a significant difference between genders. Response to the Hypothesis: The statistical data do not support Hypothesis 19 (H19). There is no statistically significant difference between the intentions of men and women to taste fresh products with life-extending stickers. Gender does not significantly influence the intention to taste these products.

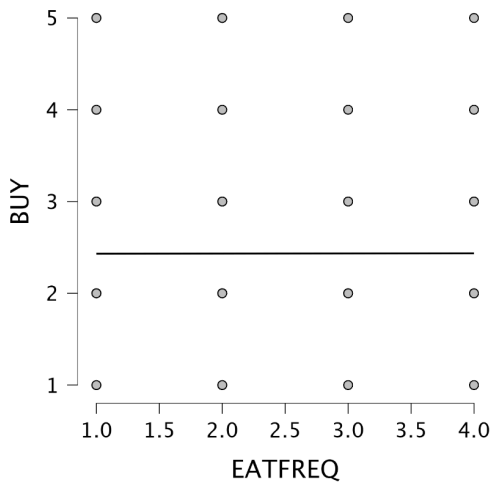
Hypothesis 30 (H30). Eating frequency has a positive impact on the intention to buy fresh products with a life-extending sticker.

Corrélations de Spearman

			Rho de Spearman	p
EATFREQ	-	BUY	0.007	.826

Scatter plots

EATFREQ vs. BUY



Hypothesis 30 is decisively refuted (not supported).

Based on the statistical analysis:

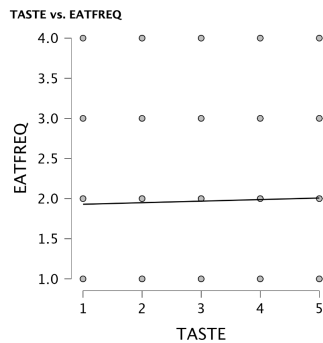
- The Spearman correlation coefficient is essentially zero ($\rho = 0.007$).
- The relationship lacks statistical significance ($p = 0.826$).

There is no statistical evidence to support the idea that "Eating frequency" has a positive impact on the "intention to buy" fresh products with a life-extending sticker.

Hypothesis 31 (H31). Eating frequency has a positive impact on the intention to taste fresh products with a life-extending sticker.

Corrélations de Spearman				
		Rho de Spearman	p	
TASTE	-	EATFREQ	0.021	.513

Scatter plots



Hypothesis 31 is refuted (not supported).

Based on the statistical analysis:

- The Spearman correlation coefficient is extremely weak ($\rho = 0.021$).
- The relationship is not statistically significant ($p = 0.513$).

There is no significant statistical evidence to support the claim that eating frequency has a positive impact on the intention to taste fresh products with a life-extending sticker.

Hypothesis 38 (H38). Age has an impact on company perception.

Test de Kruskal–Wallis

Test de Kruskal–Wallis

Facteur	Statistiques	dl	p	Classement ϵ^2	95% CI for Rank ϵ^2	
					Inférieur	Supérieur
AGE	6.183	4	.186	0.006	0.001	0.023

Hypothesis 38 is refuted (not supported).

The Kruskal-Wallis Test shows no statistical evidence that age has a significant impact on company perception ($p = 0.186$). Furthermore, the associated effect size is extremely small ($\epsilon^2 = 0.006$), confirming that any observed variation is negligible.

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